H4 EMA Crossover Quality Audit

RegimeForgeEA Research

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H4 EMA crossover quality audit

Decision

Not high quality for deployment. The selected candidate remains a useful research artifact, but it fails this stricter quality audit because independent samples are small, short-only validation is negative, and modestly adverse costs eliminate validation profitability. No live or demo activation should be inferred from the original proxy-screen pass.

Cost sensitivity

The 35-point baseline is the original fixed proxy assumption. Higher-cost rows remove the entry-spread lock only to isolate the effect of cost; they are not recommended operating settings.

Spread points	Sample	Return	Max drawdown	Trades	PF
35	Training	3.79%	2.08%	63	1.46
35	Validation	0.60%	2.17%	24	1.18
35	Holdout	0.40%	1.03%	8	1.29
70	Training	3.62%	2.11%	63	1.44
70	Validation	0.53%	2.20%	24	1.15
70	Holdout	0.39%	1.04%	8	1.28
105	Training	3.06%	2.21%	63	1.36
105	Validation	-0.09%	2.24%	24	0.98
105	Holdout	0.37%	1.05%	8	1.27
140	Training	2.89%	2.31%	63	1.34
140	Validation	-0.16%	2.28%	24	0.96
140	Holdout	-0.44%	1.62%	8	0.69

Long/short decomposition at the original cost

Side	Sample	Return	Max drawdown	Trades	PF
Long only	Training	2.82%	1.70%	35	1.65
Long only	Validation	0.77%	0.88%	15	1.40
Long only	Holdout	0.68%	0.67%	4	2.66
Short only	Training	1.52%	1.74%	34	1.34
Short only	Validation	-0.29%	1.52%	11	0.82
Short only	Holdout	0.36%	0.83%	5	1.38

Calendar-year stability at the original cost

Year	Return	Max drawdown	Trades	PF
2021	-0.28%	1.72%	18	0.89
2022	0.59%	2.07%	23	1.17
2023	3.73%	0.65%	20	3.18
2024	0.60%	2.17%	24	1.18
2025	0.40%	1.03%	8	1.29

Quality assessment

The original chronological screen is valid as a preliminary hypothesis test: the rule uses completed bars, takes both directions, defines exits before the holdout, and has positive aggregate results at the baseline proxy cost. It is not a sufficient basis for a high-quality classification. The 2024 validation contains 24 trades and the 2025 holdout only 8. Calendar results are uneven, with a negative 2021 and most profit concentrated in 2023. Most importantly, the short-only sleeve loses in validation (PF below 1.0), so the evidence for the requested short-selling component is not stable.

At 105 fixed spread points, validation turns negative; at 140 points, both validation and holdout are negative. Broker-native XAUUSD costs can be variable and can include spread, commission, swap, slippage, and gap risk, none of which are captured by the baseline model. Therefore the candidate does not clear a reasonable robustness threshold for deployment.

Required evidence before reconsideration

1. MT5 backtests on broker-native XAUUSD bid/ask data with commission, swap, realistic variable spread, and the broker's actual contract specification.
2. A pre-registered, untouched out-of-sample period with a materially larger trade count than the present eight-trade holdout.
3. Side-specific evidence showing that both long and short components are robust, or an explicit redesign to remove the unsupported side.
4. Demo forward testing with order-level reconciliation against the EA log.
5. A new quality audit using the same adverse-cost and stability criteria.